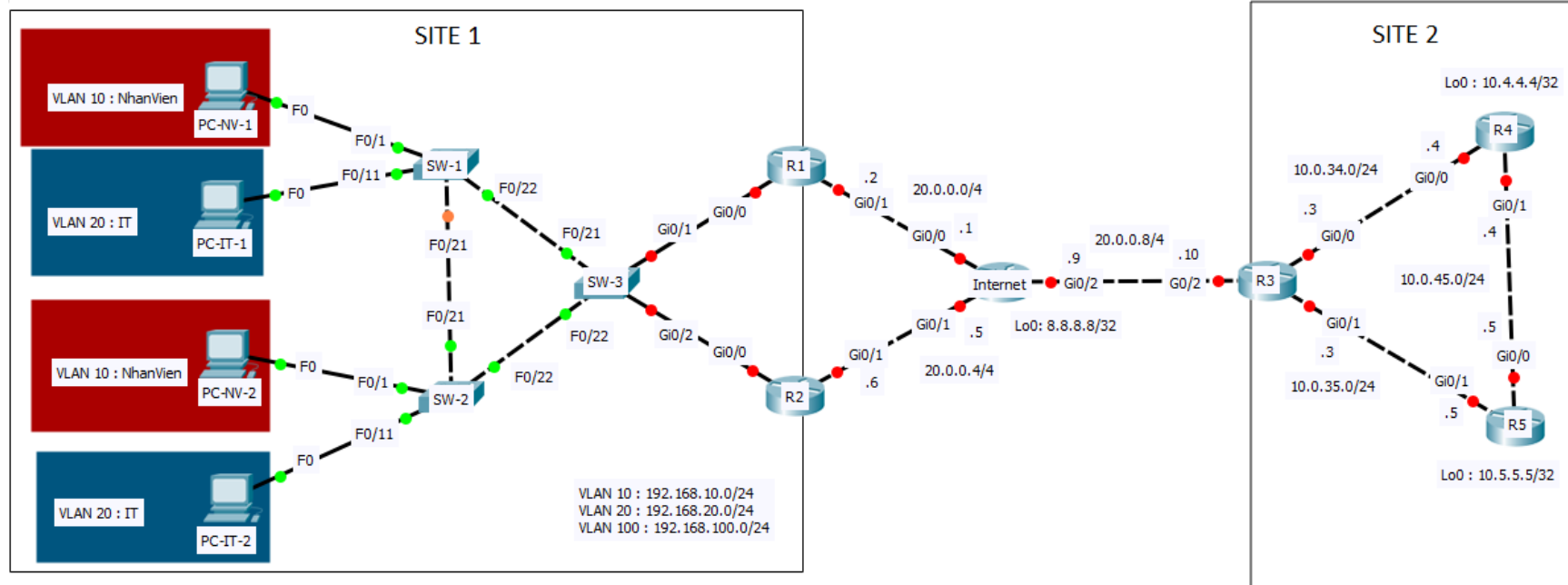


HƯỚNG DẪN LAB

LAB 29 – LAB Ôn tập

Designed by : Nguyễn Phú Thịnh

Sơ đồ LAB :



Thiết bị

Tên thiết bị	Loại thiết bị
SW-1	Switch 2960
SW-2	Switch 2960
SW-3	Switch 2960
R1	Router 2911
R2	Router 2911
R3	Router 2911
R4	Router 2911
R5	Router 2911
Internet	Router 2911

Yêu cầu

- Đấu nối và đặt tên thiết bị như sơ đồ

Đặt IP cho Router Internet :

Port/Interface	IP
Gi0/0	20.0.0.1/30
Gi0/1	20.0.0.5/30
Gi0/2	20.0.0.9/30
Loopback 0	8.8.8.8/32

Cấu hình SITE 1 :

- Tạo 3 VLAN như sau :
 - VLAN 10 : NhanVien
 - VLAN 20 : IT
 - VLAN 100 : QuanLy
 - VLAN 999 : Unused
- Cấu trunk trên các switch như sau:

Thiết bị	Port	Mode
SW-1	F0/21	Trunk
	F0/22	Trunk
SW-2	F0/21	Trunk
	F0/22	Trunk
SW-3	F0/21	Trunk
	F0/22	Trunk
	Gi0/1	Trunk

	Gi0/2	Trunk
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- Encapsulation : 802.1q
- Allowed VLAN : 10,20,100
- Native VLAN 999

- Gán các port access vào VLAN như sau :

Thiết bị	Port	VLAN
SW-1	F0/1 - 10	10
	F0/11 - 20	20
SW-2	F0/1 - 10	10
	F0/11 - 20	20

- Cấu hình STP như sau :
 - SW-3 làm root switch cho tất cả các VLAN
 - Các port access trên SW-1 và SW-2 là portfast, bật tính năng BPDU Guard
- Đặt IP cho các port/interface :

Thiết bị	Port/Interface	IP
R1	Gi0/0.10	192.168.10.1
	Gi0/0.20	192.168.20.1
	Gi0/0.100	192.168.100.1
	Gi0/1	20.0.0.2/30
R2	Gi0/0.10	192.168.10.2
	Gi0/0.20	192.168.20.2
	Gi0/0.100	192.168.100.2
	Gi0/1	20.0.0.6/30
SW-1	vlan 100	192.168.100.11
SW-2	vlan 100	192.168.100.12
SW-3	vlan 100	192.168.100.13

- Cấu hình HSRP cho VLAN 10 như sau :
 - IP ảo 192.168.10.3
 - R1 là Active (priority 105)
 - R2 là Standby (priority 100)
 - Bật tính năng preempt
 - Track interface Gi0/1 trên R1
- Cấu hình HSRP cho VLAN 20 như sau :
 - IP ảo 192.168.20.3
 - R1 là Standby (priority 100)
 - R2 là Active (priority 105)

- Bật tính năng preempt
- Track interface Gi0/1 trên R2
- Cấu hình HSRP cho VLAN 100 như sau :
 - IP ảo 192.168.100.3
 - R1 là Active (priority 105)
 - R2 là Standby (priority 100)
- Cấu hình R1 làm DHCP Server cho VLAN 10 :
 - Subnet 192.168.10.0/24
 - Default Gateway 192.168.10.3
 - DNS 8.8.8.8
- Cấu hình R2 làm DHCP Server cho VLAN 20 :
 - Subnet 192.168.20.0/24
 - Default Gateway 192.168.20.3
 - DNS 8.8.8.8
- Cấu hình default route và NAT trên hai router R1 và R2 để các PC trong VLAN 10 và VLAN 20 có thể ra được Internet
- Cấu hình telnet cho các thiết bị
 - Dùng username admin, password newstar
 - Password enable (secret) : newstar123
 - Mã hóa tất cả password
- Cấu hình Access List để chặn không cho các PC trong VLAN 10 truy cập vào VLAN 100
- Cấu hình Access List chỉ cho phép telnet vào thiết bị bằng địa chỉ IP trong VLAN 100

Cấu hình SITE 2

- Đặt IP cho các interface như sau :

Thiết bị	Port/Interface	IP
R3	Gi0/0	10.0.34.3/24
	Gi0/1	10.0.35.3/24
	Gi0/2	20.0.0.10/30
R4	Gi0/0	10.0.34.4/24
	Gi0/1	10.0.45.4/24
	Loopback 0	10.4.4.4/32
R5	Gi0/0	10.0.45.5/24
	Gi0/1	10.0.35.5/24
	Loopback 0	10.4.4.4./32

- Cấu hình các router R3, R4, R5 chạy EIGRP :
 - Quảng bá tất cả IP private (không quảng bá IP 20.0.0.X)
 - Cấu hình passive interface cho Loopback 0 trên R4 và R5
- Cấu hình default route và NAT trên R3 để các router còn lại trong SITE 2 có thể ra được Internet.
- Redistribute default route vào EIGRP để R4 và R5 tự động có default route.

Các bước thực hiện

Cấu hình Router Internet

Trên router giả lập Internet, chỉ đặt IP cho các interface

Internet					
Internet (config) # interface Gi0/0					
Internet (config-if) # ip address 20.0.0.1 255.255.255.252					
Internet (config-if) # no shut					
Internet (config-if) # exit					
Internet (config) # interface Gi0/1					
Internet (config-if) # ip address 20.0.0.5 255.255.255.252					
Internet (config-if) # no shut					
Internet (config-if) # exit					
Internet (config) # interface Gi0/2					
Internet (config-if) # ip address 20.0.0.9 255.255.255.252					
Internet (config-if) # no shut					
Internet (config-if) # exit					
Internet (config) # interface loopback 0					
Internet (config-if) # ip address 8.8.8.8 255.255.255.255					
Internet (config-if) # exit					
Kiểm tra :					
Internet# show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	20.0.0.1	YES	manual	up	down
GigabitEthernet0/1	20.0.0.5	YES	manual	up	down
GigabitEthernet0/2	20.0.0.9	YES	manual	up	down
Loopback0	8.8.8.8	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Cấu hình Site 1

Bước 1 : Tạo VLAN cho các switch

SW-1

```
SW-1(config)#vlan 10
SW-1(config-vlan)#name NhanVien
SW-1(config-vlan)#exit
SW-1(config)#vlan 20
SW-1(config-vlan)#name IT
SW-1(config-vlan)#exit
SW-1(config)#vlan 100
SW-1(config-vlan)#name QuanLy
SW-1(config-vlan)#exit
SW-1(config)#vlan 999
SW-1(config-vlan)#name Unused
SW-1(config-vlan)#exit
```

Kiểm tra :

SW-1#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	NhanVien	active	
20	IT	active	
100	QuanLy	active	
999	Unused	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-2

```
SW-2(config)#vlan 10
SW-2(config-vlan)#name NhanVien
SW-2(config-vlan)#exit
SW-2(config)#vlan 20
SW-2(config-vlan)#name IT
SW-2(config-vlan)#exit
SW-2(config)#vlan 100
SW-2(config-vlan)#name QuanLy
SW-2(config-vlan)#exit
SW-2(config)#vlan 999
SW-2(config-vlan)#name Unused
SW-2(config-vlan)#exit
```

Kiểm tra :

SW-2#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2

10	NhanVien	active
20	IT	active
100	QuanLy	active
999	Unused	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

SW-3		
<pre>SW-3(config)#vlan 10 SW-3(config-vlan)#name NhanVien SW-3(config-vlan)#exit SW-3(config)#vlan 20 SW-3(config-vlan)#name IT SW-3(config-vlan)#exit SW-3(config)#vlan 100 SW-3(config-vlan)#name QuanLy SW-3(config-vlan)#exit SW-3(config)#vlan 999 SW-3(config-vlan)#name Unused SW-3(config-vlan)#exit</pre>		
Kiểm tra : SW-3#show vlan brief		
VLAN	Name	Status
1	default	active
10	NhanVien	active
20	IT	active
100	QuanLy	active
999	Unused	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

Bước 2 : Cấu hình Trunk trên các switch

SW-1				
<pre>SW-1(config)#interface range F0/21-22 SW-1(config-if-range)#switchport mode trunk SW-1(config-if-range)#switchport trunk allowed vlan 10,20,100 SW-1(config-if-range)#switchport trunk native vlan 999</pre>				
Kiểm tra : SW-1#show interface trunk				
Port	Mode	Encapsulation	Status	Native vlan
Fa0/21	on	802.1q	trunking	999
Fa0/22	on	802.1q	trunking	999

Port	Vlans allowed on trunk
Fa0/21	10,20,100
Fa0/22	10,20,100
Port	Vlans allowed and active in management domain
Fa0/21	10,20,100
Fa0/22	10,20,100
Port	Vlans in spanning tree forwarding state and not pruned
Fa0/21	none
Fa0/22	none

SW-2	
SW-2(config)# interface range F0/21-22	
SW-2(config-if-range)# switchport mode trunk	
SW-2(config-if-range)# switchport trunk allowed vlan 10,20,100	
SW-2(config-if-range)# switchport trunk native vlan 999	
Kiểm tra :	
SW-1# show interface trunk	
Port	Mode Encapsulation Status Native vlan
Fa0/21	on 802.1q trunking 999
Fa0/22	on 802.1q trunking 999
Port	Vlans allowed on trunk
Fa0/21	10,20,100
Fa0/22	10,20,100
Port	Vlans allowed and active in management domain
Fa0/21	10,20,100
Fa0/22	10,20,100
Port	Vlans in spanning tree forwarding state and not pruned
Fa0/21	none
Fa0/22	none

SW-3	
SW-3(config)# interface range F0/21-22	
SW-3(config-if-range)# switchport mode trunk	
SW-3(config-if-range)# switchport trunk allowed vlan 10,20,100	
SW-3(config-if-range)# switchport trunk native vlan 999	
SW-3(config-if-range)# exit	
SW-3(config)# interface range Gi0/1-2	
SW-3(config-if-range)# switchport mode trunk	
SW-3(config-if-range)# switchport trunk allowed vlan 10,20,100	
SW-3(config-if-range)# switchport trunk native vlan 999	
Kiểm tra :	
SW-3# show interface trunk	
Port	Mode Encapsulation Status Native vlan
Fa0/21	on 802.1q trunking 999
Fa0/22	on 802.1q trunking 999
Port	Vlans allowed on trunk
Fa0/21	10,20,100
Fa0/22	10,20,100
Port	Vlans allowed and active in management domain
Fa0/21	10,20,100
Fa0/22	10,20,100

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/21	10,20,100
Fa0/22	10,20,100

Lưu ý : có thể lúc này port Gi0/1 và Gi0/2 chưa xuất hiện vì đầu xa của các port đang kết nối với 2 router và đang bị shutdown. Ở các bước tiếp theo, sau khi cấu hình router, 2 port này sẽ xuất hiện.

Bước 3 : Gán các port access vào VLAN

SW-1			
SW-1(config)# interface range F0/1-10			
SW-1(config-if-range)# switchport mode access			
SW-1(config-if-range)# switchport access vlan 10			
SW-1(config-if-range)# exit			
SW-1(config)# interface range F0/11-20			
SW-1(config-if-range)# switchport mode access			
SW-1(config-if-range)# switchport access vlan 20			
Kiểm tra :			
SW-1# show vlan brief			
VLAN	Name	Status	Ports
1	default	active	Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	NhanVien	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
20	IT	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20
100	QuanLy	active	
999	Unused	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-2			
SW-2(config)# interface range F0/1-10			
SW-2(config-if-range)# switchport mode access			
SW-2(config-if-range)# switchport access vlan 10			
SW-2(config-if-range)# exit			
SW-2(config)# interface range F0/11-20			
SW-2(config-if-range)# switchport mode access			
SW-2(config-if-range)# switchport access vlan 20			
SW-2# show vlan brief			
VLAN	Name	Status	Ports
1	default	active	Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	NhanVien	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
20	IT	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18

		Fa0/19, Fa0/20
100	QuanLy	active
999	Unused	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

Bước 4 : Cấu hình portfast trên các port access

SW-1
SW-1(config)# interface range F0/1-20 SW-1(config-if-range)# spanning-tree portfast SW-1(config-if-range)# spanning-tree bpduguard enable

SW-2
SW-2(config)# interface range F0/1-20 SW-2(config-if-range)# spanning-tree portfast SW-2(config-if-range)# spanning-tree bpduguard enable

Bước 5 : Cấu hình để SW-3 làm root switch cho tất cả VLAN

Do chỉ có 3 VLAN được qua trunk là 10,20 và 100 nên chỉ cần cấu hình để SW-3 làm root switch cho 3 VLAN này .

SW-3
SW-3(config)# spanning-tree vlan 10 root primary SW-3(config)# spanning-tree vlan 20 root primary SW-3(config)# spanning-tree vlan 100 root primary

Bước 6 : Đặt IP cho các interface

R1
R1(config)# interface Gi0/0 R1(config-if)# no shutdown R1(config-if)# exit R1(config)# interface Gi0/0.10 R1(config-subif)# encapsulation dot1Q 10 R1(config-subif)# ip address 192.168.10.1 255.255.255.0 R1(config-subif)# exit R1(config)# interface Gi0/0.20 R1(config-subif)# encapsulation dot1Q 20 R1(config-subif)# ip add 192.168.20.1 255.255.255.0 R1(config-subif)# exit R1(config)# interface Gi0/0.100 R1(config-subif)# encapsulation dot1Q 100 R1(config-subif)# ip address 192.168.100.1 255.255.255.0 R1(config-subif)# exit R1(config)# interface Gi0/1 R1(config-if)# ip address 20.0.0.2 255.255.255.252 R1(config-if)# no shut

Kiểm tra :

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.100	192.168.100.1	YES	manual	up	up
GigabitEthernet0/1	20.0.0.2	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

R2

R2(config)#interface Gi0/0

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#interface Gi0/0.10

R2(config-subif)#encapsulation dot1Q 10

R2(config-subif)#ip address 192.168.10.2 255.255.255.0

R2(config-subif)#exit

R2(config)#interface Gi0/0.20

R2(config-subif)#encapsulation dot1Q 20

R2(config-subif)#ip address 192.168.20.2 255.255.255.0

R2(config-subif)#exit

R2(config)#interface Gi0/0.100

R2(config-subif)#encapsulation dot1Q 100

R2(config-subif)#ip address 192.168.100.2 255.255.255.0

R2(config-subif)#exit

R2(config)#interface Gi0/1

R2(config-if)#ip address 20.0.0.6 255.255.255.252

R2(config-if)#no shut

Kiểm tra :

R2#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.2	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.2	YES	manual	up	up
GigabitEthernet0/0.100	192.168.100.2	YES	manual	up	up
GigabitEthernet0/1	20.0.0.6	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

SW-1

SW-1(config)#interface vlan 100

SW-1(config-if)#ip address 192.168.100.11 255.255.255.0

SW-1(config-if)#no shut

Kiểm tra :

SW-1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	down	down
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down

FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	up	up
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	up	up
FastEthernet0/22	unassigned	YES	manual	up	up
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	unassigned	YES	manual	administratively down	down
Vlan100	192.168.100.11	YES	manual	up	up

SW-2					
SW-2 (config) #interface vlan 100					
SW-2 (config-if) #ip address 192.168.100.12 255.255.255.0					
SW-2 (config-if) #no shut					
Kiểm tra :					
SW-2#show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	down	down
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	up	up
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	up	up
FastEthernet0/22	unassigned	YES	manual	up	up
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	unassigned	YES	manual	administratively down	down
Vlan100	192.168.100.12	YES	manual	up	up

SW-3					
SW-3 (config) #interface vlan 100					

```
SW-3(config-if)#ip address 192.168.100.13 255.255.255.0
SW-3(config-if)#no shut
```

Kiểm tra :

```
SW-3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	down	down
FastEthernet0/2	unassigned	YES	manual	down	down
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	up	up
FastEthernet0/22	unassigned	YES	manual	up	up
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	manual	up	up
Vlan1	unassigned	YES	manual	administratively down	down
Vlan100	192.168.100.13	YES	manual	up	up

Bước 7 : Cấu hình HSRP

R1

```
R1(config)#interface Gi0/0.10
R1(config-subif)#standby 1 ip 192.168.10.3
R1(config-subif)#standby 1 priority 105
R1(config-subif)#standby 1 preempt
R1(config-subif)#standby 1 track Gi0/1
R1(config-subif)#exit
R1(config)#interface Gi0/0.20
R1(config-subif)#standby 2 ip 192.168.20.3
R1(config-subif)#standby 2 preempt
R1(config-subif)#exit
R1(config)#interface Gi0/0.100
R1(config-subif)#standby 3 ip 192.168.100.3
R1(config-subif)#standby 3 priority 105
R1(config-subif)#standby 3 preempt
```

R2

```
R2(config)#interface Gi0/0.10
R2(config-subif)#standby 1 ip 192.168.10.3
R2(config-subif)#standby 1 preempt
R2(config-subif)#exit
```

```

R2(config)#interface Gi0/0.20
R2(config-subif)#standby 2 ip 192.168.20.3
R2(config-subif)#standby 2 priority 105
R2(config-subif)#standby 2 preempt
R2(config-subif)#standby 2 track Gi0/1
R2(config-subif)#exit
R2(config)#interface Gi0/0.100
R2(config-subif)#standby 3 ip 192.168.100.3
R2(config-subif)#standby 3 preempt

```

Bước 8 : Kiểm tra lại cấu hình HSRP

R1

```

R1#show standby
GigabitEthernet0/0.10 - Group 1
  State is Active
    8 state changes, last state change 02:02:06
  Virtual IP address is 192.168.10.3
  Active virtual MAC address is 0000.0C07.AC01
    Local virtual MAC address is 0000.0C07.AC01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.585 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.10.2
  Priority 105 (configured 105)
    Track interface GigabitEthernet0/1 state Up decrement 10
  Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 2
  State is Standby
    11 state changes, last state change 02:05:52
  Virtual IP address is 192.168.20.3
  Active virtual MAC address is 0000.0C07.AC02
    Local virtual MAC address is 0000.0C07.AC02 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.703 secs
  Preemption enabled
  Active router is 192.168.20.2, priority 105 (expires in 6 sec)
    MAC address is 0000.0C07.AC02
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
  State is Active
    6 state changes, last state change 02:02:52
  Virtual IP address is 192.168.100.3
  Active virtual MAC address is 0000.0C07.AC03
    Local virtual MAC address is 0000.0C07.AC03 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.817 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.2
  Priority 105 (configured 105)
  Group name is hsrp-Gig-3 (default)

```

R2

```

R2#show standby
GigabitEthernet0/0.10 - Group 1

```

```

State is Standby
  3 state changes, last state change 02:05:22
Virtual IP address is 192.168.10.3
Active virtual MAC address is 0000.0C07.AC01
  Local virtual MAC address is 0000.0C07.AC01 (v1 default)
Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.063 secs
Preemption enabled
Active router is 192.168.10.1
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 2
State is Active
  3 state changes, last state change 02:05:34
Virtual IP address is 192.168.20.3
Active virtual MAC address is 0000.0C07.AC02
  Local virtual MAC address is 0000.0C07.AC02 (v1 default)
Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.751 secs
Preemption enabled
Active router is local
Standby router is 192.168.20.1, priority 100 (expires in 7 sec)
Priority 105 (configured 105)
  Track interface GigabitEthernet0/1 state Up decrement 10
Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
State is Standby
  3 state changes, last state change 02:06:21
Virtual IP address is 192.168.100.3
Active virtual MAC address is 0000.0C07.AC03
  Local virtual MAC address is 0000.0C07.AC03 (v1 default)
Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.099 secs
Preemption enabled
Active router is 192.168.100.1
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig-3 (default)

```

Bước 9 : Cấu hình DHCP Server

Lưu ý : tạo 2 pool trên cả hai router

R1
<pre> R1(config)#ip dhcp pool NhanVien R1(dhcp-config)#network 192.168.10.0 255.255.255.0 R1(dhcp-config)#default-router 192.168.10.3 R1(dhcp-config)#dns 8.8.8.8 R1(dhcp-config)#exit R1(config)#ip dhcp pool IT R1(dhcp-config)#network 192.168.20.0 255.255.255.0 R1(dhcp-config)#default-router 192.168.20.3 R1(dhcp-config)#dns 8.8.8.8 R1(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10 R1(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10 </pre>

R2

```
R2(config)#ip dhcp pool NhanVien
R2(dhcp-config)#network 192.168.10.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.10.3
R2(dhcp-config)#dns 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp pool IT
R2(dhcp-config)#network 192.168.20.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.20.3
R2(dhcp-config)#dns 8.8.8.8
R2(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
R2(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
```

Bước 10 : Cho PC nhận IP bằng DHCP

PC-NV-1

The screenshot shows the configuration window for PC-NV-1. The 'Config' tab is selected, and the 'IP Configuration' sub-tab is active. Under 'IP Configuration', the 'DHCP' radio button is selected, and a message 'DHCP request successful.' is displayed. The fields for IP Address, Subnet Mask, Default Gateway, and DNS Server are populated with the values 192.168.10.11, 255.255.255.0, 192.168.10.3, and 8.8.8.8 respectively. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address field is empty, the Link Local Address field contains 'FE80::260:47FF:FE33:7DC8', and the IPv6 Gateway and IPv6 DNS Server fields are also empty. A 'Top' button is located at the bottom left of the window.

PC-IT-1

PC-IT-1

Physical Config Desktop Programming Attributes

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.20.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.3

DNS Server 8.8.8.8

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FE4B:7B1C

IPv6 Gateway

IPv6 DNS Server

☐ Top

PC-NV-2

PC-NV-2

Physical Config Desktop Programming Attributes

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.10.12

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.3

DNS Server 8.8.8.8

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

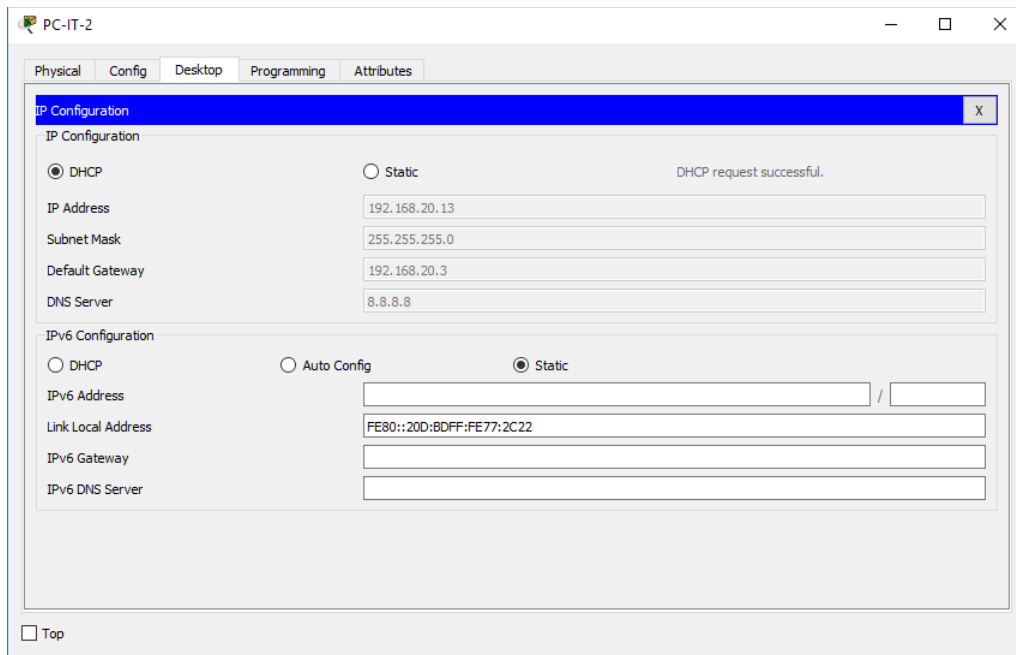
Link Local Address FE80::2D0:FFFF:FEA3:9E70

IPv6 Gateway

IPv6 DNS Server

☐ Top

PC-IT-2



Lưu ý : IP nhận có thể khác, nhưng subnet/subnet mask, default gateway và dns server phải giống

Bước 11 : Cấu hình default route trên hai router

```

R1
R1(config)#ip route 0.0.0.0 0.0.0.0 Gi0/1

Kiểm tra :
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/30 is directly connected, GigabitEthernet0/1
L       20.0.0.2/32 is directly connected, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
    192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, GigabitEthernet0/0.100
L       192.168.100.1/32 is directly connected, GigabitEthernet0/0.100
S*    0.0.0.0/0 is directly connected, GigabitEthernet0/1

R1#ping 8.8.8.8

```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

R2

```
R2(config)#ip route 0.0.0.0 0.0.0.0 Gi0/1
```

Kiểm tra :

```
R2#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
```

```
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.4/30 is directly connected, GigabitEthernet0/1
L       20.0.0.6/32 is directly connected, GigabitEthernet0/1
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.2/32 is directly connected, GigabitEthernet0/0.10
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.2/32 is directly connected, GigabitEthernet0/0.20
      192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, GigabitEthernet0/0.100
L       192.168.100.2/32 is directly connected, GigabitEthernet0/0.100
S*    0.0.0.0/0 is directly connected, GigabitEthernet0/1
```

```
R2#ping 8.8.8.8
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

Bước 12 : Cấu hình NAT

R1

```
R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#access-list 1 permit 192.168.20.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface Gi0/1 overload
R1(config)#interface Gi0/0.10
R1(config-subif)#ip nat inside
R1(config-subif)#exit
R1(config)#interface Gi0/0.20
R1(config-subif)#ip nat inside
R1(config-subif)#exit
R1(config)#interface Gi0/1
R1(config-if)#ip nat outside
```

Kiểm tra :

```
R1#ping
Protocol [ip]:
Target IP address: 8.8.8.8
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: GigabitEthernet0/0.10
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 192.168.10.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

```
R1#ping
Protocol [ip]:
Target IP address: 8.8.8.8
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: GigabitEthernet0/0.20
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 192.168.20.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

R2

```
R2(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R2(config)#access-list 1 permit 192.168.20.0 0.0.0.255
R2(config)#ip nat inside source list 1 interface Gi0/1 overload
R2(config)#interface Gi0/0.10
R2(config-subif)#ip nat inside
R2(config-subif)#exit
R2(config)#interface Gi0/0.20
R2(config-subif)#ip nat inside
R2(config-subif)#exit
R2(config)#interface Gi0/1
R2(config-if)#ip nat outside
```

Kiểm tra :

```
R2#ping
Protocol [ip]:
Target IP address: 8.8.8.8
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
```

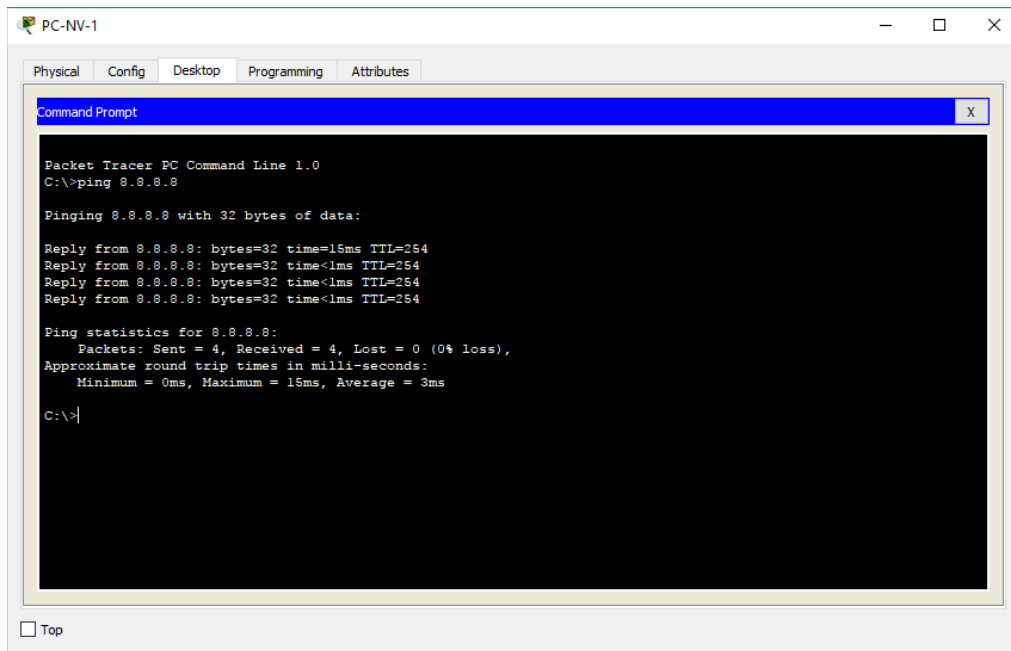
```
Source address or interface: GigabitEthernet0/0.10
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 192.168.10.2
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R2#ping
Protocol [ip]:
Target IP address: 8.8.8.8
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: GigabitEthernet0/0.20
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 192.168.20.2
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/3 ms
```

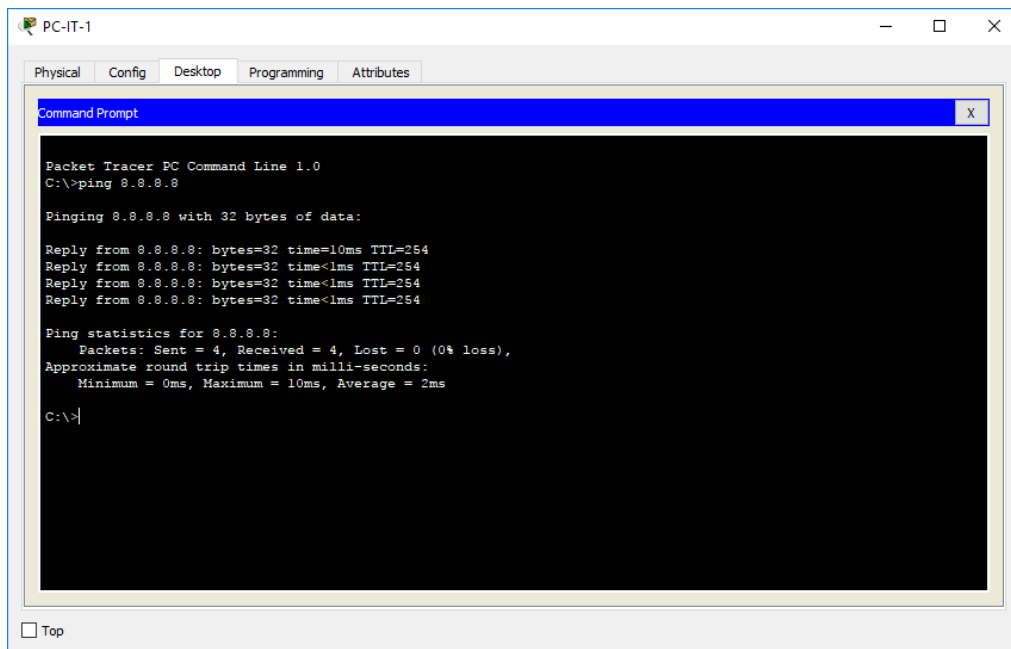
Bước 13 : Kiểm tra kết nối Internet từ PC

Trên từng PC, thực hiện lệnh ping 8.8.8.8

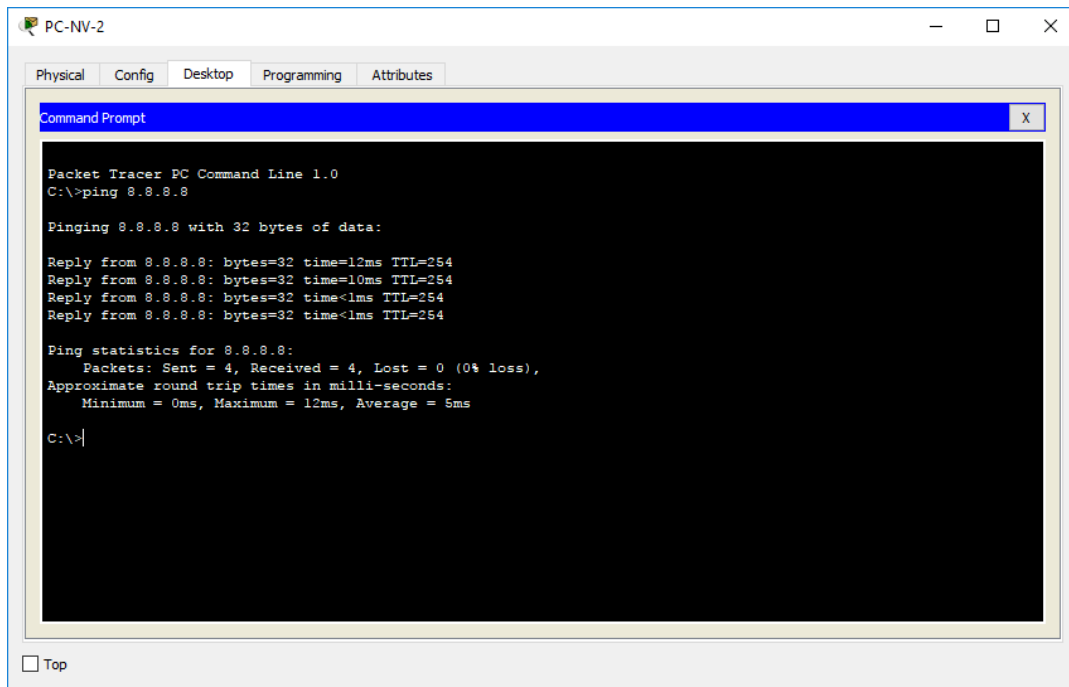
PC-NV-1



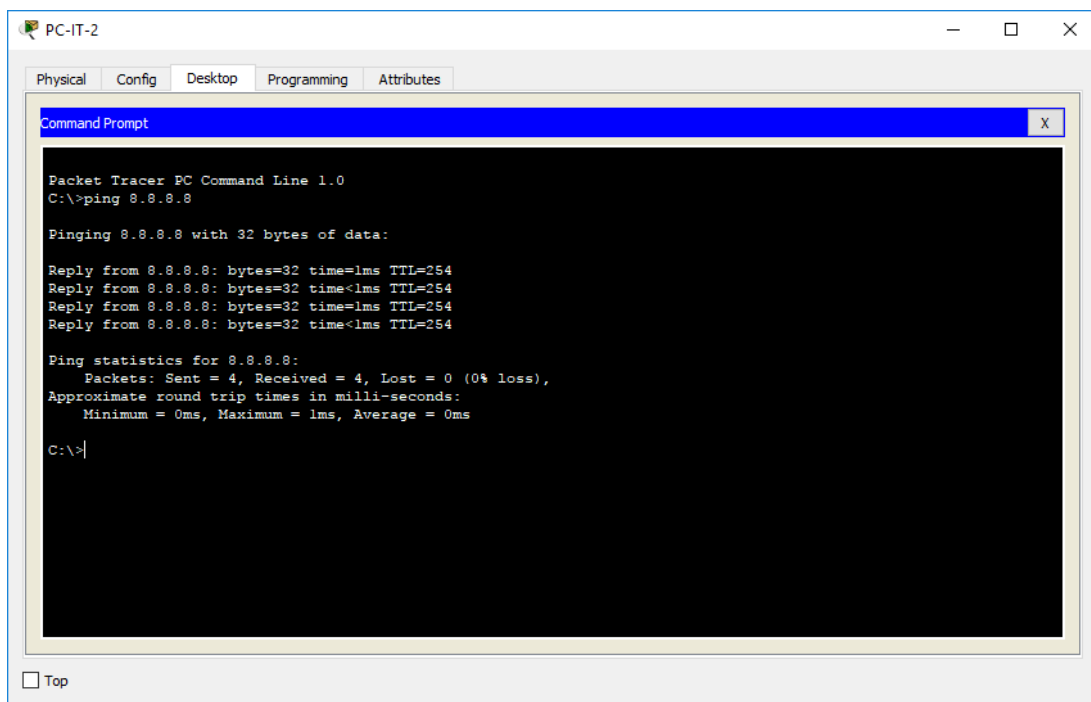
PC-IT-1



PC-NV-2



PC-IT-2



Bước 14 : Kiểm tra tính dự phòng cho VLAN 10

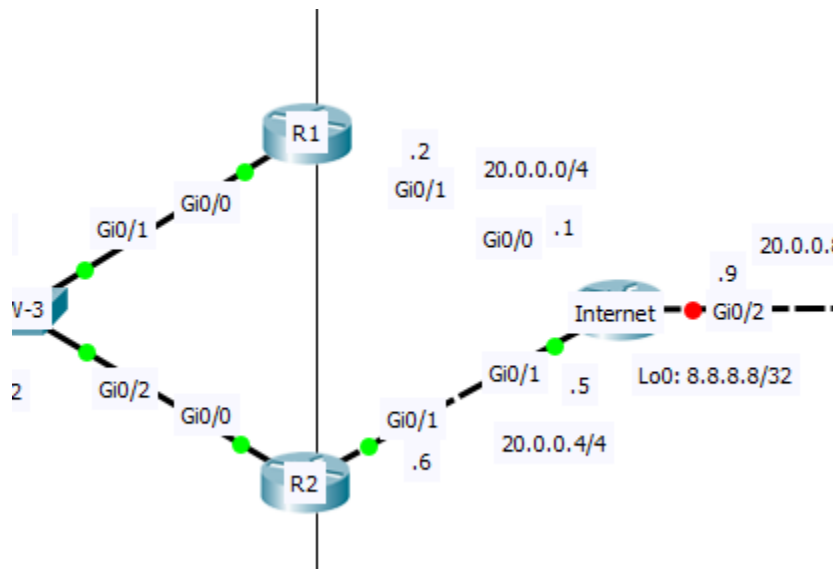
Trên **PC-NV-1**, ping liên tục đến 8.8.8.8 bằng lệnh **ping -t 8.8.8.8**

```
C:\>ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=3ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
```

Trong lúc ping, ngắt kết nối Internet của R1



Kết quả : ping gián đoạn 1 khoảng thời gian ngắn, sau đó vẫn ping bình thường

```
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=10ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
```

Router R1 đã trở thành Standby cho vlan 10, priority bị trừ chỉ còn 95.

R1
<pre>R1#show standby GigabitEthernet0/0.10 - Group 1 State is Standby 12 state changes, last state change 02:44:50 Virtual IP address is 192.168.10.3 Active virtual MAC address is 0000.0C07.AC01 Local virtual MAC address is 0000.0C07.AC01 (v1 default)</pre>


```

Hello time 3 sec, hold time 10 sec
  Next hello sent in 0.989 secs
Preemption enabled
Active router is 192.168.10.2, priority 100 (expires in 8 sec)
  MAC address is 0000.0C07.AC01
Standby router is local
Priority 95 (configured 105)
  Track interface GigabitEthernet0/1 state Up decrement 10
  Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 2
  State is Standby
    11 state changes, last state change 02:05:52
  Virtual IP address is 192.168.20.3
  Active virtual MAC address is 0000.0C07.AC02
    Local virtual MAC address is 0000.0C07.AC02 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.241 secs
  Preemption enabled
  Active router is 192.168.20.2, priority 105 (expires in 9 sec)
    MAC address is 0000.0C07.AC02
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
  State is Active
    6 state changes, last state change 02:02:52
  Virtual IP address is 192.168.100.3
  Active virtual MAC address is 0000.0C07.AC03
    Local virtual MAC address is 0000.0C07.AC03 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.521 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.2
  Priority 105 (configured 105)
  Group name is hsrp-Gig-3 (default)

```

Router R2 đã trở thành Active cho vlan 10

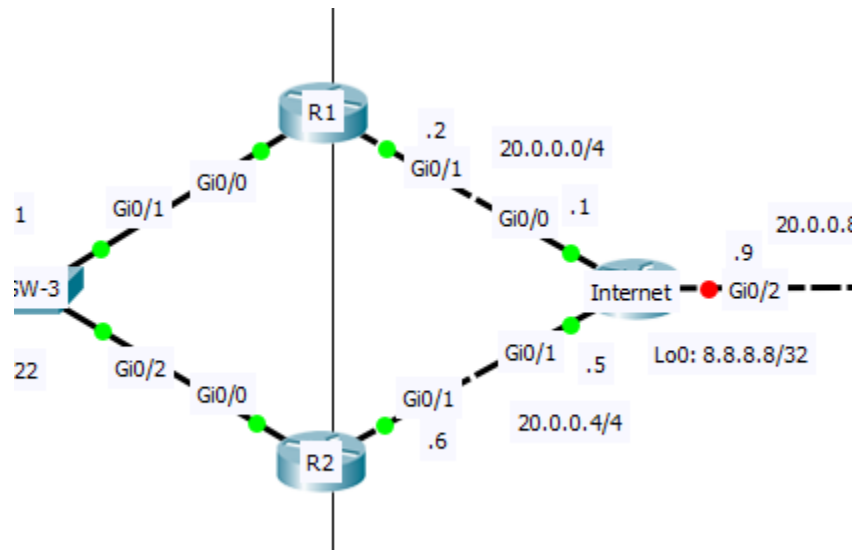
R2
<pre> R2#show standby GigabitEthernet0/0.10 - Group 1 State is Active 4 state changes, last state change 02:44:31 Virtual IP address is 192.168.10.3 Active virtual MAC address is 0000.0C07.AC01 Local virtual MAC address is 0000.0C07.AC01 (v1 default) Hello time 3 sec, hold time 10 sec Next hello sent in 2.851 secs Preemption enabled Active router is local Standby router is 192.168.10.1, priority 95 (expires in 6 sec) Priority 100 (default 100) Group name is hsrp-Gig-1 (default) GigabitEthernet0/0.20 - Group 2 State is Active 3 state changes, last state change 02:05:34 Virtual IP address is 192.168.20.3 Active virtual MAC address is 0000.0C07.AC02 Local virtual MAC address is 0000.0C07.AC02 (v1 default) Hello time 3 sec, hold time 10 sec Next hello sent in 1.65 secs </pre>

```

Preemption enabled
Active router is local
Standby router is 192.168.20.1, priority 100 (expires in 6 sec)
Priority 105 (configured 105)
  Track interface GigabitEthernet0/1 state Up decrement 10
Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
State is Standby
  3 state changes, last state change 02:06:21
Virtual IP address is 192.168.100.3
Active virtual MAC address is 0000.0C07.AC03
  Local virtual MAC address is 0000.0C07.AC03 (v1 default)
Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.602 secs
Preemption enabled
Active router is 192.168.100.1
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig-3 (default)

```

Khôi phục lại kết nối Internet cho R1



Kết quả : ping gián đoạn 1 khoảng thời gian ngắn, sau đó vẫn ping bình thường

```

Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254
Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=13ms TTL=254
Reply from 8.8.8.8: bytes=32 time=12ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254

```

Priority của R1 được khôi phục lại 105, R1 giành lại quyền Active

R1
<pre> R1#show standby GigabitEthernet0/0.10 - Group 1 State is Active 14 state changes, last state change 02:48:33 Virtual IP address is 192.168.10.3 </pre>

```
Active virtual MAC address is 0000.0C07.AC01
  Local virtual MAC address is 0000.0C07.AC01 (v1 default)
Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.231 secs
Preemption enabled
Active router is local
Standby router is 192.168.10.2, priority 100 (expires in 2806061 sec)
Priority 105 (configured 105)
  Track interface GigabitEthernet0/1 state Up decrement 10
  Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 2
  State is Standby
    11 state changes, last state change 02:05:52
  Virtual IP address is 192.168.20.3
  Active virtual MAC address is 0000.0C07.AC02
    Local virtual MAC address is 0000.0C07.AC02 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.186 secs
  Preemption enabled
  Active router is 192.168.20.2, priority 105 (expires in 3 sec)
    MAC address is 0000.0C07.AC02
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
  State is Active
    6 state changes, last state change 02:02:52
  Virtual IP address is 192.168.100.3
  Active virtual MAC address is 0000.0C07.AC03
    Local virtual MAC address is 0000.0C07.AC03 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.619 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.1
  Priority 105 (configured 105)
  Group name is hsrp-Gig-3 (default)
```

Bước 15 : Kiểm tra tính dự phòng cho vlan 20

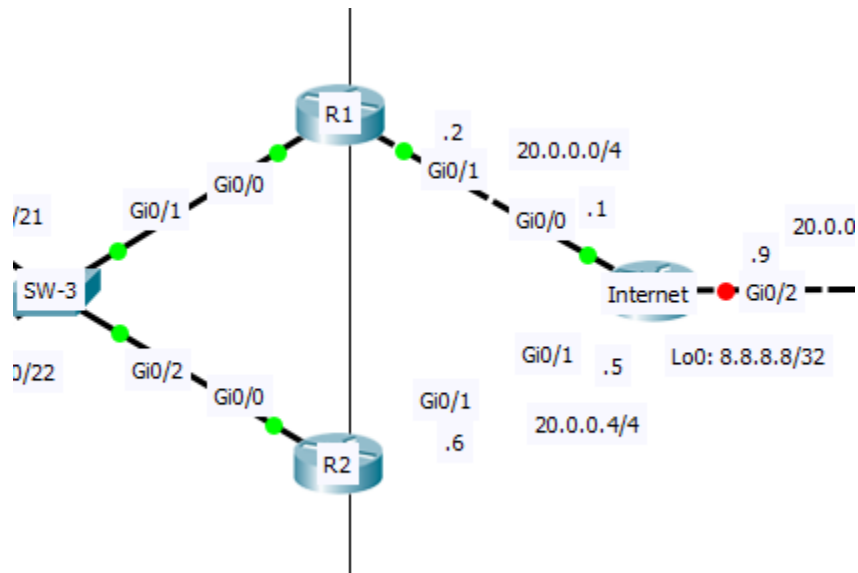
Trên **PC-IT-1**, ping liên tục đến 8.8.8.8 bằng lệnh **ping -t 8.8.8.8**

```
C:\>ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
Reply from 8.8.8.8: bytes=32 time<lms TTL=254
```

Trong lúc ping, ngắt kết nối Internet của **R2**



Kết quả : ping gián đoạn 1 khoảng thời gian ngắn, sau đó vẫn ping bình thường

```

Reply from 8.8.8.8: bytes=32 time=10ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 192.168.20.2: Destination host unreachable.
Reply from 192.168.20.2: Destination host unreachable.
Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254

```

Router R2 đã trở thành Standby cho vlan 20, priority bị trừ chỉ còn 95.

R2
<pre> R2#show standby GigabitEthernet0/0.10 - Group 1 State is Standby 6 state changes, last state change 00:00:49 Virtual IP address is 192.168.10.3 Active virtual MAC address is 0000.0C07.AC01 Local virtual MAC address is 0000.0C07.AC01 (v1 default) Hello time 3 sec, hold time 10 sec Next hello sent in 2.378 secs Preemption enabled Active router is 192.168.10.1 Standby router is local Priority 100 (default 100) Group name is hsrp-Gig-1 (default) GigabitEthernet0/0.20 - Group 2 State is Standby 9 state changes, last state change 00:02:45 Virtual IP address is 192.168.20.3 Active virtual MAC address is 0000.0C07.AC02 Local virtual MAC address is 0000.0C07.AC02 (v1 default) Hello time 3 sec, hold time 10 sec Next hello sent in 1.252 secs Preemption enabled Active router is 192.168.20.1, priority 100 (expires in 8 sec) </pre>

```
MAC address is 0000.0C07.AC02
Standby router is local
Priority 95 (configured 105)
Track interface GigabitEthernet0/1 state Up decrement 10
Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
State is Standby
6 state changes, last state change 00:00:50
Virtual IP address is 192.168.100.3
Active virtual MAC address is 0000.0C07.AC03
Local virtual MAC address is 0000.0C07.AC03 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.896 secs
Preemption enabled
Active router is 192.168.100.1, priority 105 (expires in 8 sec)
MAC address is 0000.0C07.AC03
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig-3 (default)
```

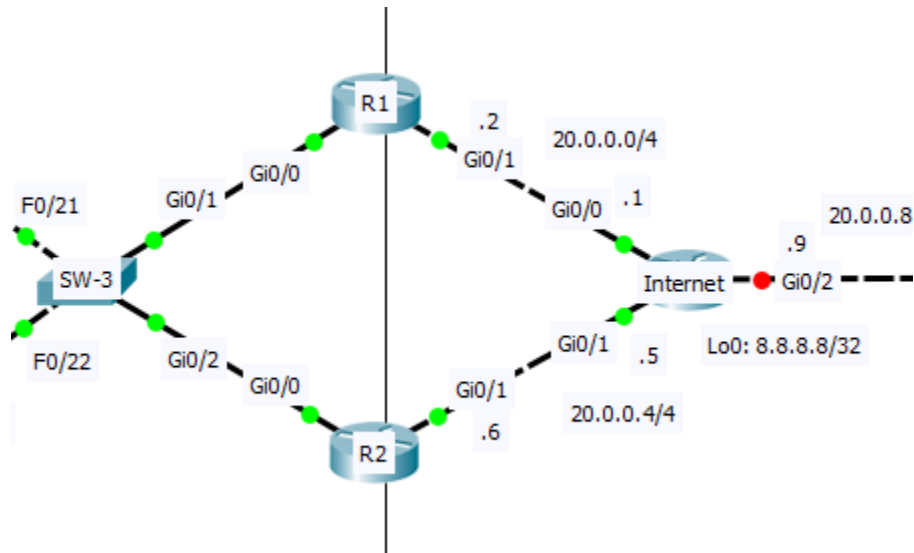
Router R1 đã trở thành Active cho vlan 20

R1

```
R1#show standby
GigabitEthernet0/0.10 - Group 1
State is Active
5 state changes, last state change 00:00:37
Virtual IP address is 192.168.10.3
Active virtual MAC address is 0000.0C07.AC01
Local virtual MAC address is 0000.0C07.AC01 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.372 secs
Preemption enabled
Active router is local
Standby router is 192.168.10.2
Priority 105 (configured 105)
Track interface GigabitEthernet0/1 state Up decrement 10
Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 2
State is Active
7 state changes, last state change 00:02:27
Virtual IP address is 192.168.20.3
Active virtual MAC address is 0000.0C07.AC02
Local virtual MAC address is 0000.0C07.AC02 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.857 secs
Preemption enabled
Active router is local
Standby router is 192.168.20.2, priority 95 (expires in 6 sec)
Priority 100 (default 100)
Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
State is Active
5 state changes, last state change 00:00:29
Virtual IP address is 192.168.100.3
Active virtual MAC address is 0000.0C07.AC03
Local virtual MAC address is 0000.0C07.AC03 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.657 secs
Preemption enabled
Active router is local
Standby router is 192.168.100.2, priority 100 (expires in 6 sec)
```

```
Priority 105 (configured 105)  
Group name is hsrp-Gig-3 (default)
```

Khôi phục lại kết nối Internet cho R2



Kết quả : ping gián đoạn 1 khoảng thời gian ngắn, sau đó vẫn ping bình thường

```
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254  
Request timed out.  
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254  
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
```

Priority của R2 được khôi phục lại 105, R2 giành lại quyền Active

R1

R2#show standby

```
GigabitEthernet0/0.10 - Group 1  
  State is Standby  
    6 state changes, last state change 00:00:49  
  Virtual IP address is 192.168.10.3  
  Active virtual MAC address is 0000.0C07.AC01  
    Local virtual MAC address is 0000.0C07.AC01 (v1 default)  
  Hello time 3 sec, hold time 10 sec  
    Next hello sent in 0.518 secs  
  Preemption enabled  
  Active router is 192.168.10.1  
  Standby router is local  
  Priority 100 (default 100)  
  Group name is hsrp-Gig-1 (default)  
GigabitEthernet0/0.20 - Group 2  
  State is Active  
    11 state changes, last state change 00:04:41  
  Virtual IP address is 192.168.20.3  
  Active virtual MAC address is 0000.0C07.AC02  
    Local virtual MAC address is 0000.0C07.AC02 (v1 default)  
  Hello time 3 sec, hold time 10 sec
```

```

Next hello sent in 0.918 secs
Preemption enabled
Active router is local
Standby router is 192.168.20.1, priority 100 (expires in 8 sec)
Priority 105 (configured 105)
Track interface GigabitEthernet0/1 state Up decrement 10
Group name is hsrp-Gig-2 (default)
GigabitEthernet0/0.100 - Group 3
State is Standby
6 state changes, last state change 00:00:50
Virtual IP address is 192.168.100.3
Active virtual MAC address is 0000.0C07.AC03
Local virtual MAC address is 0000.0C07.AC03 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 2.412 secs
Preemption enabled
Active router is 192.168.100.1, priority 105 (expires in 6 sec)
MAC address is 0000.0C07.AC03
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig-3 (default)

```

Bước 16 : cấu hình telnet cho các thiết bị

SW-1

```

SW-1(config)#enable secret newstar123
SW-1(config)#username admin password newstar
SW-1(config)#service password-encryption
SW-1(config)#line vty 0 4
SW-1(config-line)#login local
SW-1(config-line)#exit
SW-1(config)#ip default-gateway 192.168.100.3

```

SW-2

```

SW-2(config)#enable secret newstar123
SW-2(config)#username admin password newstar
SW-2(config)#service password-encryption
SW-2(config)#line vty 0 4
SW-2(config-line)#login local
SW-2(config-line)#exit
SW-2(config)#ip default-gateway 192.168.100.3

```

SW-3

```

SW-3(config)#enable secret newstar123
SW-3(config)#username admin password newstar
SW-3(config)#service password-encryption
SW-3(config)#line vty 0 4
SW-3(config-line)#login local
SW-3(config-line)#exit
SW-3(config)#ip default-gateway 192.168.100.3

```

R1

```

R1(config)#enable secret newstar123
R1(config)#username admin password newstar
R1(config)#service password-encryption
R1(config)#line vty 0 4

```

```
R1(config-line)#login local
```

R2

```
R2(config)#enable secret newstar123  
R2(config)#username admin password newstar  
R2(config)#service password-encryption  
R2(config)#line vty 0 4  
R2(config-line)#login local
```

Bước 17 : Telnet thử vào các thiết bị từ PC

Ví dụ, telnet vào **R1** từ **PC-NV-1**

```
C:\>telnet 192.168.100.1  
Trying 192.168.100.1 ...Open  
  
User Access Verification  
  
Username: admin  
  
Password:  
R1>  
R1>enable  
Password:  
R1#  
R1#
```

telnet vào **R2** từ **PC-NV-1**

```
C:\>telnet 192.168.100.2  
Trying 192.168.100.2 ...Open  
  
User Access Verification  
  
Username: admin  
  
Password:  
R2>  
R2>en  
Password:  
R2#  
R2#
```

telnet vào **SW-1** từ **PC-NV-1**

```
C:\>telnet 192.168.100.11  
Trying 192.168.100.11 ...Open  
  
User Access Verification  
  
Username: admin  
  
Password:  
SW-1>  
SW-1>en  
Password:  
SW-1#  
SW-1#
```

telnet vào **SW-2** từ **PC-NV-1**


```

C:\>telnet 192.168.100.12
Trying 192.168.100.12 ...Open

User Access Verification

Username: admin

Password:
SW-2>en
Password:
SW-2#

```

telnet vào SW-3 từ PC-NV-1

```

C:\>telnet 192.168.100.13
Trying 192.168.100.13 ...Open

User Access Verification

Username: admin

Password:
SW-3>
SW-3>enable
Password:
SW-3#

```

Bước 18 : Cấu hình Access List chỉ cho VLAN 20 truy cập vào VLAN 100

R1
<pre> R1(config)#ip access-list extended DENY-TO-VL100 R1(config-ext-nacl)#deny ip any 192.168.100.0 0.0.0.255 R1(config-ext-nacl)#permit ip any any R1(config-ext-nacl)#exit R1(config)#interface Gi0/0.10 R1(config-subif)#ip access-group DENY-TO-VL100 in </pre>

R2
<pre> R2(config)#ip access-list extended DENY-TO-VL100 R2(config-ext-nacl)#deny ip any 192.168.100.0 0.0.0.255 R2(config-ext-nacl)#permit ip any any R2(config-ext-nacl)#exit R2(config)#interface Gi0/0.10 R2(config-subif)#ip access-group DENY-TO-VL100 in </pre>

Bước 19 : Kiểm tra ACL

Trên PC-NV-1, lần lượt ping 8.8.8.8, 192.168.20.1 (IP trong VLAN 20), 192.168.100.1 (IP trong VLAN 100)

```

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=11ms TTL=254

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms

```

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.100.1

Pinging 192.168.100.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.100.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Kết quả :

- VLAN 10 có thể ra internet
- VLAN 10 có thể trao đổi dữ liệu với VLAN 20
- VLAN 10 không thể truy cập vào VLAN 100

Bước 20 : cấu hình access list chỉ cho phép telnet vào thiết bị bằng IP trong VLAN 100

R1
R1(config)# ip access-list extended TELNET-to-VL100-ONLY
R1(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.100.0 0.0.0.255 eq 23
R1(config-ext-nacl)# deny tcp any any eq 23
R1(config-ext-nacl)# permit ip any any
R1(config)# interface Gi0/0.20
R1(config-subif)# ip access-group TELNET-to-VL100-ONLY in

R2
R2(config)# ip access-list extended TELNET-to-VL100-ONLY
R2(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.100.0 0.0.0.255 eq 23
R2(config-ext-nacl)# deny tcp any any eq 23
R2(config-ext-nacl)# permit ip any any
R2(config)# interface Gi0/0.20
R2(config-subif)# ip access-group TELNET-to-VL100-ONLY in

Bước 21 : Kiểm tra Access List

Trên PC-IT-1, lần lượt telnet vào R1 bằng các IP 192.168.10.1 (IP trong vlan10), 192.168.20.1 (IP trong vlan 20), 192.168.100.1 (IP trong vlan 100)

```

C:\>telnet 192.168.10.1
Trying 192.168.10.1 ...
% Connection timed out; remote host not responding
C:\>
C:\>telnet 192.168.20.1
Trying 192.168.20.1 ...
% Connection timed out; remote host not responding
C:\>
C:\>telnet 192.168.100.1
Trying 192.168.100.1 ...Open

User Access Verification

Username: admin

Password:
R1>
R1>en
Password:
R1#

```

Kết quả : chỉ có thể telnet vào R1 bằng IP 192.168.100.1 (IP trong VLAN 100)

Cấu hình SITE 2

Bước 1 : Đặt địa chỉ IP cho các interface

R3					
R3(config)# interface Gi0/0					
R3(config-if)# ip address 10.0.34.3 255.255.255.0					
R3(config-if)# no shut					
R3(config-if)# exit					
R3(config)# interface Gi0/1					
R3(config-if)# ip address 10.0.35.3 255.255.255.0					
R3(config-if)# no shut					
R3(config-if)# exit					
R3(config)# interface Gi0/2					
R3(config-if)# ip address 20.0.0.10 255.255.255.252					
R3(config-if)# no shut					
Kiểm tra :					
R3# show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.34.3	YES	manual	up	down
GigabitEthernet0/1	10.0.35.3	YES	manual	up	down
GigabitEthernet0/2	20.0.0.10	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R4					
R4(config)# interface Gi0/0					
R4(config-if)# ip address 10.0.34.4 255.255.255.0					
R4(config-if)# no shut					
R4(config-if)# exit					
R4(config)# interface Gi0/1					
R4(config-if)# ip address 10.0.45.4 255.255.255.0					
R4(config-if)# no shut					
R4(config-if)# exit					
R4(config)# interface loopback 0					
R4(config-if)# ip address 10.4.4.4 255.255.255.255					

Kiểm tra :

R4#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.34.4	YES	manual	up	up
GigabitEthernet0/1	10.0.45.4	YES	manual	up	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Loopback0	10.4.4.4	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R5

```
R5(config)#interface Gi0/0
R5(config-if)#ip address 10.0.45.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#interface Gi0/1
R5(config-if)#ip address 10.0.35.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#interface loopback 0
R5(config-if)#ip address 10.5.5.5 255.255.255.255
```

Kiểm tra :

R5#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.45.5	YES	manual	up	up
GigabitEthernet0/1	10.0.35.5	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Loopback0	10.5.5.5	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Bước 2 : Cấu hình EIGRP cho các router

R3

```
R3(config)#router eigrp 65000
R3(config-router)#network 10.0.34.0 0.0.0.255
R3(config-router)#network 10.0.35.0 0.0.0.255
R3(config-router)#no auto-summary
```

Kiểm tra :

R3#show ip protocols

```
Routing Protocol is "eigrp 65000 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 65000
  EIGRP-IPv4 Protocol for AS(65000)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.34.3
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1
```

```
Automatic Summarization: disabled
Automatic address summarization:
Maximum path: 4
Routing for Networks:
  10.0.34.0/24
  10.0.35.0/24
Routing Information Sources:
  Gateway      Distance      Last Update
Distance: internal 90 external 170
```

R4

```
R4(config)#router eigrp 65000
R4(config-router)#network 10.0.34.0 0.0.0.255
R4(config-router)#network 10.0.45.0 0.0.0.255
R4(config-router)#network 10.4.4.4 0.0.0.0
R4(config-router)#no auto-summary
R4(config-router)#passive-interface loopback 0
```

Kiểm tra :

```
R4#show ip protocols
```

```
Routing Protocol is "eigrp 65000 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 65000
  EIGRP-IPv4 Protocol for AS(65000)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.34.4
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1
```

```
Automatic Summarization: disabled
Automatic address summarization:
Maximum path: 4
Routing for Networks:
  10.0.34.0/24
  10.0.45.0/24
  10.4.4.4/32
Passive Interface(s):
  Loopback0
Routing Information Sources:
  Gateway      Distance      Last Update
  10.0.34.3     90            8074223
Distance: internal 90 external 170
```

R5

```
R5(config)#router eigrp 65000
R5(config-router)#network 10.0.45.0 0.0.0.255
R5(config-router)#network 10.0.35.5 0.0.0.255
R5(config-router)#network 10.5.5.5 0.0.0.0
R5(config-router)#passive-interface loopback 0
R5(config-router)#no auto-summary
```

Kiểm tra :

R5#**show ip protocols**

```
Routing Protocol is "eigrp 65000 "  
  Outgoing update filter list for all interfaces is not set  
  Incoming update filter list for all interfaces is not set  
  Default networks flagged in outgoing updates  
  Default networks accepted from incoming updates  
  Redistributing: eigrp 65000  
  EIGRP-IPv4 Protocol for AS(65000)  
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0  
    NSF-aware route hold timer is 240  
    Router-ID: 10.0.35.5  
    Topology : 0 (base)  
      Active Timer: 3 min  
      Distance: internal 90 external 170  
      Maximum path: 4  
      Maximum hopcount 100  
      Maximum metric variance 1  
  
  Automatic Summarization: disabled  
  Automatic address summarization:  
  Maximum path: 4  
  Routing for Networks:  
    10.0.45.0/24  
    10.0.35.0/24  
    10.5.5.5/32  
  Passive Interface(s):  
    Loopback0  
  Routing Information Sources:  
    Gateway         Distance      Last Update  
    10.0.45.4        90           8208991  
    10.0.35.3        90           8214850  
  Distance: internal 90 external 170
```

Bước 3 : Kiểm tra bảng định tuyến

R3

R3#**show ip route**

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
       * - candidate default, U - per-user static route, o - ODR  
       P - periodic downloaded static route  
  
Gateway of last resort is not set  
  
  10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks  
C       10.0.34.0/24 is directly connected, GigabitEthernet0/0  
L       10.0.34.3/32 is directly connected, GigabitEthernet0/0  
C       10.0.35.0/24 is directly connected, GigabitEthernet0/1  
L       10.0.35.3/32 is directly connected, GigabitEthernet0/1  
D       10.0.45.0/24 [90/3072] via 10.0.34.4, 00:04:45, GigabitEthernet0/0  
          [90/3072] via 10.0.35.5, 00:02:31, GigabitEthernet0/1  
D       10.4.4.4/32 [90/130816] via 10.0.34.4, 00:04:37, GigabitEthernet0/0  
D       10.5.5.5/32 [90/130816] via 10.0.35.5, 00:02:23, GigabitEthernet0/1  
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C       20.0.0.8/30 is directly connected, GigabitEthernet0/2
```

```
L      20.0.0.10/32 is directly connected, GigabitEthernet0/2
```

R4

```
R4#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.34.0/24 is directly connected, GigabitEthernet0/0
L       10.0.34.4/32 is directly connected, GigabitEthernet0/0
D       10.0.35.0/24 [90/3072] via 10.0.34.3, 00:05:28, GigabitEthernet0/0
          [90/3072] via 10.0.45.5, 00:03:09, GigabitEthernet0/1
C       10.0.45.0/24 is directly connected, GigabitEthernet0/1
L       10.0.45.4/32 is directly connected, GigabitEthernet0/1
C       10.4.4.4/32 is directly connected, Loopback0
D       10.5.5.5/32 [90/130816] via 10.0.45.5, 00:03:00, GigabitEthernet0/1
```

R5

```
R5#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D       10.0.34.0/24 [90/3072] via 10.0.45.4, 00:03:54, GigabitEthernet0/0
          [90/3072] via 10.0.35.3, 00:03:48, GigabitEthernet0/1
C       10.0.35.0/24 is directly connected, GigabitEthernet0/1
L       10.0.35.5/32 is directly connected, GigabitEthernet0/1
C       10.0.45.0/24 is directly connected, GigabitEthernet0/0
L       10.0.45.5/32 is directly connected, GigabitEthernet0/0
D       10.4.4.4/32 [90/130816] via 10.0.45.4, 00:03:54, GigabitEthernet0/0
C       10.5.5.5/32 is directly connected, Loopback0
```

Bước 4 : Cấu hình default route trên R3

R3

```
R3(config)#ip route 0.0.0.0 0.0.0.0 Gi0/2
```

Kiểm tra :

```
R3#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
```

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

```
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C    10.0.34.0/24 is directly connected, GigabitEthernet0/0
L    10.0.34.3/32 is directly connected, GigabitEthernet0/0
C    10.0.35.0/24 is directly connected, GigabitEthernet0/1
L    10.0.35.3/32 is directly connected, GigabitEthernet0/1
D    10.0.45.0/24 [90/3072] via 10.0.34.4, 00:07:14, GigabitEthernet0/0
      [90/3072] via 10.0.35.5, 00:05:00, GigabitEthernet0/1
D    10.4.4.4/32 [90/130816] via 10.0.34.4, 00:07:06, GigabitEthernet0/0
D    10.5.5.5/32 [90/130816] via 10.0.35.5, 00:04:52, GigabitEthernet0/1
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.8/30 is directly connected, GigabitEthernet0/2
L    20.0.0.10/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/2
```

R3#

R3#**ping 8.8.8.8**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

.!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Bước 5 : Cấu hình NAT trên R3

R3

```
R3(config)#access-list 1 permit 10.0.0.0 0.255.255.255
R3(config)#ip nat inside source list 1 interface Gi0/2
R3(config)#interface Gi0/0
R3(config-if)#ip nat inside
R3(config-if)#exit
R3(config)#interface Gi0/1
R3(config-if)#ip nat inside
R3(config-if)#exit
R3(config)#interface Gi0/2
R3(config-if)#ip nat outside
```

Kiểm tra :

R3#**ping**

Protocol [ip]:

Target IP address: **8.8.8.8**

Repeat count [5]:

Datagram size [100]:

Timeout in seconds [2]:

Extended commands [n]: **y**

Source address or interface: **GigabitEthernet0/0**

Type of service [0]:

Set DF bit in IP header? [no]:

Validate reply data? [no]:

Data pattern [0xABCD]:

Loose, Strict, Record, Timestamp, Verbose[none]:

Sweep range of sizes [n]:

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

Packet sent with a source address of 10.0.34.3


```

!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R3#ping
Protocol [ip]:
Target IP address: 8.8.8.8
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: GigabitEthernet0/1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 10.0.35.3
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/2 ms

```

Bước 6 : Redistribute default route

R3
<pre> R3(config)#router eigrp 65000 R3(config-router)#redistribute static metric 100000 1 255 1 1500 Kiểm tra : R3#show ip eigrp topology IP-EIGRP Topology Table for AS 65000/ID(20.0.0.10) Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply, r - Reply status P 0.0.0.0/0, 1 successors, FD is 25856 via Rstatic (25856/0) P 10.0.34.0/24, 1 successors, FD is 2816 via Connected, GigabitEthernet0/0 P 10.0.35.0/24, 1 successors, FD is 2816 via Connected, GigabitEthernet0/1 P 10.0.45.0/24, 2 successors, FD is 3072 via 10.0.34.4 (3072/2816), GigabitEthernet0/0 via 10.0.35.5 (3072/2816), GigabitEthernet0/1 P 10.4.4.4/32, 1 successors, FD is 130816 via 10.0.34.4 (130816/128256), GigabitEthernet0/0 P 10.5.5.5/32, 1 successors, FD is 130816 via 10.0.35.5 (130816/128256), GigabitEthernet0/1 </pre>

Bước 7 : Kiểm tra bảng định tuyến trên R4 và R5

R4
<pre> R4#show ip route Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 </pre>

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.0.34.3 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C 10.0.34.0/24 is directly connected, GigabitEthernet0/0
L 10.0.34.4/32 is directly connected, GigabitEthernet0/0
D 10.0.35.0/24 [90/3072] via 10.0.34.3, 00:13:23, GigabitEthernet0/0
[90/3072] via 10.0.45.5, 00:11:04, GigabitEthernet0/1
C 10.0.45.0/24 is directly connected, GigabitEthernet0/1
L 10.0.45.4/32 is directly connected, GigabitEthernet0/1
C 10.4.4.4/32 is directly connected, Loopback0
D 10.5.5.5/32 [90/130816] via 10.0.45.5, 00:10:55, GigabitEthernet0/1
D*EX 0.0.0.0/0 [170/26112] via 10.0.34.3, 00:01:34, GigabitEthernet0/0

R4#ping 8.8.8.8

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/20 ms

R4#ping

Protocol [ip]:

Target IP address: 8.8.8.8

Repeat count [5]:

Datagram size [100]:

Timeout in seconds [2]:

Extended commands [n]: y

Source address or interface: Loopback0

Type of service [0]:

Set DF bit in IP header? [no]:

Validate reply data? [no]:

Data pattern [0xABCD]:

Loose, Strict, Record, Timestamp, Verbose[none]:

Sweep range of sizes [n]:

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

Packet sent with a source address of 10.4.4.4

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R5

R5#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is 10.0.35.3 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D 10.0.34.0/24 [90/3072] via 10.0.45.4, 00:13:01, GigabitEthernet0/0
[90/3072] via 10.0.35.3, 00:12:55, GigabitEthernet0/1
C 10.0.35.0/24 is directly connected, GigabitEthernet0/1
L 10.0.35.5/32 is directly connected, GigabitEthernet0/1

```
C    10.0.45.0/24 is directly connected, GigabitEthernet0/0
L    10.0.45.5/32 is directly connected, GigabitEthernet0/0
D    10.4.4.4/32 [90/130816] via 10.0.45.4, 00:13:01, GigabitEthernet0/0
C    10.5.5.5/32 is directly connected, Loopback0
D*EX 0.0.0.0/0 [170/26112] via 10.0.35.3, 00:03:26, GigabitEthernet0/1
```

R5#

R5#**ping 8.8.8.8**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R5#

R5#**ping**

Protocol [ip]:

Target IP address: **8.8.8.8**

Repeat count [5]:

Datagram size [100]:

Timeout in seconds [2]:

Extended commands [n]: **y**

Source address or interface: **Loopback0**

Type of service [0]:

Set DF bit in IP header? [no]:

Validate reply data? [no]:

Data pattern [0xABCD]:

Loose, Strict, Record, Timestamp, Verbose[none]:

Sweep range of sizes [n]:

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:

Packet sent with a source address of 10.5.5.5

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms